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Advanced Power Management Strategies for Complex Hybrid-Electric Aircraft

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Aircraft electrification for propulsion is a promising way to alleviate the negative environmental impact of conventional carbon-powered aviation. Inclusion of the electrical powertrain aims to enhance design freedom, allowing for more efficient power systems and operational schemes. In this work, a design space exploration is performed, aiming to derive power management guidelines based on aircraft environmental performance. A 19-passenger commuter aircraft employing the series/parallel partial hybrid-electric architecture is examined. Two underwing-mounted turboprop engines are combined with a boundary layer ingestion fan mounted in the aft of the aircraft and powered by an electrical drive. The primary electrical energy source is a battery system. A multidisciplinary framework is utilized, comprising modeling approaches for multipoint thermal engine design, physics-based electrical component sizing and performance, aircraft sizing, mission design, and environmental assessment. The investigation revealed that the reference designed hybrid-electric configuration with entry-into-service (EIS) 2035 assumed technologies yields roughly 18% improvement in block consumption and emissions, but an 8% increase in maximum takeoff weight (MTOW), compared to its 2014 conventional counterpart. The design space exploration for an optimal power management scheme indicated a minimum average ratio of 1:1.35 between cruise and design point hybridization power. However, even the optimally operated hybrid aircraft showcases worse environmental performance compared to the conventional design of same entry-into-service date. The investigation has revealed that the complex powertrain and hybrid architecture selected may be more suitable for larger class aircraft, where aircraft requirements can be relaxed and higher degrees of electrification are not penalized or confined by set constraints.
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1 Introduction

Ambitious goals for the reduction of aviation's environmental impact have been shared by several institutions through the past two decades [1,2]. Electrification, utilization of hydrogen, and sustainable aviation fuels all pose potential solutions to the climate change caused by flight [3]. The current work investigates electrification of a small commuter aircraft.

Several studies have been conducted to evaluate the environmental performance of regional flight. Pastra et al. [4] investigated a regional turboprop hybrid-electric configuration with entry-into-service (EIS) 2050, revealing block fuel consumption benefits in the order of 5% compared to its conventional counterpart. For a similar configuration and EIS, Cinar et al. [5] investigated the potential benefits of electrification assuming battery technology projections of 3 kWh/kg. Block fuel savings of 25% were reported for designs with degree of hybridization reaching 30%. Diamantidou et al. [6] explored the impact of range and battery technology for commuter and regional aircraft. For degrees of block energy hybridization of 25%, block fuel benefits of 25–35% were reported, while the importance of battery mass in the overall objective space was highlighted. In a comparison between configurations of EIS 2015 and 2030, Cai et al. [7] reported relative block fuel savings of 30% and 35% for the commuter and regional class, respectively. Extending the investigation of electrification benefits to a 180-passenger aircraft, Pohl et al. [8] revealed a potential power specific fuel consumption (PWSFC) improvement of 1% for each 10% in degrees of hybridization. Finally, Iwanizki et al. [9] examined several novel concepts with boundary layer ingestion fans in hybrid-electric aircraft. The study concluded that for the short range, the novel concepts offered only an average of 3% relative block fuel improvement, while they were associated with significant mass penalties.

Several investigations have been conducted to pinpoint the environmental benefits of electrification in the regional and commuter class. Nevertheless, the potential of the complex architecture of a series/parallel partial hybrid-electric aircraft with an electrically driven boundary layer ingestion aft fan remains relatively untapped. Despite the increased complexity, the introduction of electrification and boundary layer fan results in increased flexibility and several feasible modes of operation. The current work investigates this configuration with the scope of determining the space of optimal power management strategies for maximum environmental benefits.

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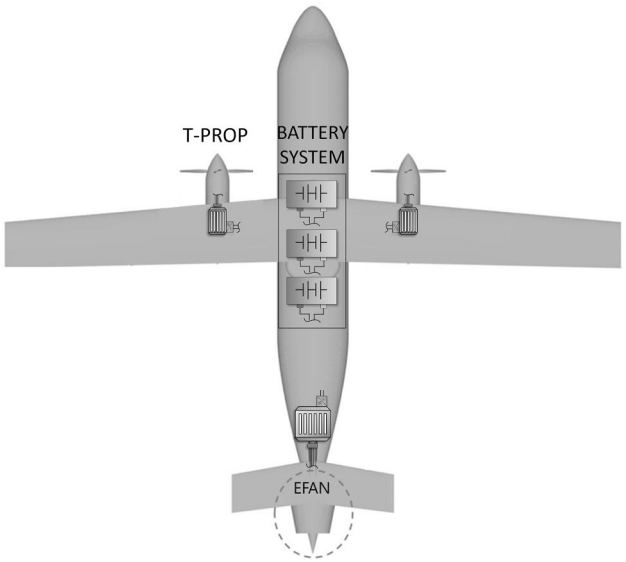


Fig. 1 Series/parallel partial hybrid-electric aircraft configuration

2 Methodology

A 19-passenger aircraft is selected as the vessel for this investigation. The reference conventional aircraft designs carry two wing-mounted turboshaft engines coupled with respective propellers. For the hybrid aircraft configurations examined, a rear fuselage-mounted electrically driven ducted fan (EFAN) is added in the propulsion system. The selected series/parallel hybrid architecture allows for different fuel-powered, hybrid, and turboelectric modes of operation, offering great flexibility and potential for environmental benefits through electrification and optimal power management strategy. The emerging hybrid aircraft configuration is shown in Fig. 1.

2.1 Thermal Powerplant. A dedicated engine model is developed to design and evaluate the performance of the thermal powerplant for the investigated aircraft configuration. The model is built within the aeroengine conceptual design tool EVA (Environmental Assessment Tool) [10] and integrated within a multidisciplinary design framework. The thermal powerplant architecture of the hybrid configuration with EIS 2035 consists of two underwing-mounted turboprop engines and a rear fuselage-mounted electrically driven boundary layer ingestion fan (EFAN).

As described by Bermperis et al. [11], a 2.5-spool geared turboprop configuration is selected with an all-axial seven-stage gas generator compressor (GGC), an axial cooled two-stage gas generator turbine (GGT), and an axial uncooled three-stage free power turbine (FPT), which is coupled to a single-speed propeller through a two-stage reduction gearbox. Moreover, a low-pressure ratio single-stage ducted EFAN is directly coupled to an electric motor (EM). The EFAN operation under the presence of boundary layer ingestion is estimated through reduced incoming Mach numbers, and variable geometry is possible in critical operating points without operability issues [12]. The whole fan operates inside the boundary layer which is roughly 1.5 times the height of the rotor blade [13].

An EIS-driven approach is deployed to design the aforementioned thermal powerplant with EIS 2035, considering material and technology improvements over the years based on publicly available engine data. The design of conventional thermal powerplants with EIS 2014 and EIS 2035 are preceded in order to provide a consistent gas generator design for the hybrid with EIS 2035. A multipoint synthesis approach and design point matching scheme have been applied to all EIS engine variants.

The pairs of target and state variables for the conventional configuration with EIS 2014 are shown in Table 1. Certification

Table 1 Performance matching scheme targets and states for the conventional configuration with EIS 2014

Targets	States
GGT metal temperature at TO	GGT cooling flow at TOC
Turbine interstage temperature at TOC	Fuel-to-air ratio at TOC
GG nozzle PR at TOC	GG inlet mass flow at TOC
Propeller loading at TOC	Propeller diameter

datasheets of existing engines with similar power levels and EIS [14] provide an interstage turbine temperature of 1050 K, corrected by end-of-life assumptions such as measurement uncertainty and degradation. Furthermore, the single-speed propeller is designed with diameter-based propeller loading of 300 kW/m^2, as suggested by NASA reports [15–17]. The evolution of turbine material capability and the suggested metal temperature trend of 3 K/yr by Kyprianidis [18] indicate a single-stage GGT metal temperature level (for stator and rotor) of 1140 K at hot-day end-of-runway takeoff conditions. Also, a fully expanded nozzle shall be ensured to reflect on the operation of a turboprop exhaust system.

An enhanced performance matching scheme is selected for the conventional and hybrid configurations with EIS 2035, which is included in Table 2. Metal temperatures for the two-stage GGT are set to 1200 K, based on the material maturity assumptions [18]. A parametric cycle design study was carried out to finally determine cruise overall pressure ratio (OPR) and T4 levels at 15 and 1400 K, respectively. This choice ensured acceptable GGC last stage blade height, and favorable cruise specific fuel consumption as well as turboprop dry weight. Propeller diameter and rotational speed (fixed throughout the flight envelope) are decided based on the choice of setting the top-of-climb propeller loading at 200 kW/m^2 and takeoff propeller tip speed at 200 m/s (due to noise regulations). The multipoint matching scheme ensures that all targets are achieved, and only feasible powertrain designs are generated.

Moving from the conventional to the hybrid configuration with EIS 2035, the cycle design remains fixed to enable a consistent comparison. The split of total power output between the propeller and the EFAN drives the slight modifications of the performance matching scheme. Specifically, EFAN mass flow is decided based on the prescribed EFAN power levels. Setting the top-level requirements for total power output while prescribing the EFAN power levels results in an indirect target of propeller power output that determines the propeller diameter.

2.2 Electrical Powerplant. Individual models and a comprehensive analysis of the underlying dependencies and tradeoffs for sizing and performance of electrical components and their power system integration are presented in detail by Bermperis et al. [11]. A briefing of the developed and utilized methodology follows. For the sizing of permanent magnet synchronous machines, an analytical methodology is developed. Primary design characteristics are selected initially, and a sinusoidal airgap flux is assumed [19,20]. The machine is designed to operate near saturation limit during peak

Table 2 Performance matching scheme targets and states for the conventional and hybrid configuration with EIS 2035

Targets	States
GGT metal temperature at TO	GGT cooling flow at TOC
OPR at CR	GGC pressure ratio at TOC
T4 at CR	T4 at TOC
Propeller tip speed at TO	Propeller rotational speed at TOC
Propeller loading at TOC	Propeller diameter
(Propeller power output at TOC, for hybrid EIS2035)	
GG nozzle PR at TOC	GG inlet mass flow at TOC
EFAN power at TOC (only for hybrid EIS 2035)	EFAN inlet mass flow at TOC (only for hybrid EIS 2035)

requirements, which leads to a minimal material requirement for the stator and rotor sizing [21–24]. The assumed electric motor material is silicon steel. Selection of cooling architecture closes the design loop, and a thermal resistance equivalent model is used to monitor temperatures during operation. For electric motor performance estimation, three loss generating phenomena are modeled [24,25]. Copper losses are calculated analytically for given machine dimensions and operating condition. Core losses are captured through a semi-empirical equation based on the fundamental electrical frequency. Finally, mechanical and windage losses are found in relation to rotational speed for given machine dimensions. Copper temperature limits, rotor tip speed, and core magnetic saturation are monitored throughout operation to ensure proper machine behavior. For the power electronic components, design is initialized at module level. A lumped thermal model is combined with sizing equations fitted to publicly available data for state-of-the-art components and scaled for the projected EIS year [26,27]. Packaging and cooling surface masses are accounted for in the final sizing outcome. Performance of power electronic components is estimated in a time-average manner capturing conduction and switching losses [28]. Equivalent models are used depending on the type of electronic converter. Direct current (DC) power cables and their insulation are sized according to ground application, and scaled to flight conditions, by catering for the impact of air density and ambient temperature [29]. The sizing and performance model of the battery refers to Li-ion cells. Cell characteristics are scaled to pack-level accounting for auxiliary components and electronic connections. Heat exchange surfaces are designed to ensure battery operation under advised limits. A resistance–capacitance equivalent model is combined with a scaled Shepherd model for the battery system performance [30], allowing for representation of both charging and discharging conditions. The packaging factor for a battery system is estimated according to assumptions for auxiliary components, cooling surfaces, and box dimensions.

The integrated electrical power system (EPS) of the examined hybrid architecture comprises of three distinct parts. A battery system and DC bus bar span the fuselage and act as an anchor point where all electrical power branches connect. The EM–EFAN subsystem refers to the electrical branch that is coupled to the aft-fan (EFAN) and drives it. An axial-flux permanent magnet synchronous motor is selected and cooled via an external cooling jacket with fins. The electric motor is coupled with a SiC DC/AC (alternating current) inverter for control. The GEN–GT (generator–gas turbine) subsystem refers to the electrical branch that is coupled to the gas turbine. A radial-flux permanent magnet synchronous machine is mechanically connected to the FPT and is used for power extraction, therefore acting as a generator. Direct winding heat exchange is considered for the generator. A SiC AC/DC rectifier controls the generator. A DC/DC converter connects the branch and DC bus bar. For both branches, power is transmitted through aluminum DC power cables. A water-based propylene glycerol solution is the selected EPS coolant.

Models for sizing and performance estimation of electrical power system components are integrated in a common framework. Details about the selected hybrid architecture are incorporated within the framework logic. Assumptions derived from the projected technology and fixed design aspects act as the design scheme's starting point. Given a mission profile and hybridization strategy, a power and shaft speed profile are derived for the electrical drives that are connected to turbomachinery components. Power demand is passed through components of respective branches accounting for performance of individual components. Operating current and voltage are also estimated for each component. The EM–EFAN and GEN–GT branches are sized, and their performance is estimated. An iterative algorithm is created to ensure that battery system outputs converge to the calculated demand of the electrical branches. Actual battery system design relies on the duration of all mission segments. The framework delivers a fully sized electrical power system, with its performance predicted at all representative mission phases and with individual component maps available.

2.3 Aircraft Design and Mission. Aircraft design and mission calculations are made within the EVA tool [10,31] and are linked to the thermal and electrical powertrain models in a multidisciplinary framework. A predefined nominal aircraft dataset is used for dimension modeling of any new aircraft configuration [32]. Sizing and analysis of each airframe component is performed to reach an initial aircraft weight estimation [32–34]. Calculations include aircraft weight breakdown throughout mission, with operating empty weight, zero fuel weight, maximum takeoff weight (MTOW), and ramp weight, all being calculated. Aircraft aerodynamics are modeled according to principles described by Jenkinson et al. [32]. Aircraft drag polar is predicted during the mission for individual components based on aircraft geometry and high lift device settings [32–34]. For the hybrid configurations, the influence of the boundary layer ingestion (BLI) fan on drag polar is accounted for through fan nacelle drag and fuselage-fan interactions [35,36]. A rubberized aircraft wing model is used to capture the impact of different powertrain designs, weights, and installation locations on aircraft maximum takeoff weight [31]. For each EIS group of aircraft designs, wing loading is fixed at takeoff and set as the wing sizing condition. A maximum lift coefficient (CLMAX) value is also fixed, while the capability of the wing design and high-lift devices to achieve CLMAX is explored and verified by Nasoulis et al. [36,37]. When a new hybrid or conventional design is created during the design space exploration, wings are also resized accordingly to ensure that new lift requirements are met. The tailplane is designed with a principle of keeping constant area ratio with the wing, while stability and acceptable static margin are always ensured in the generated designs. These rules are generated separately for the conventional and hybrid nominal aircraft given the addition of the battery system in the fuselage and the EFAN in the aircraft rear. These added components are placed strategically in the fuselage to ensure stability and a reasonable change in the aircraft's center of gravity [38,39].

Aircraft performance modeling is based on methods described by Jenkinson et al. [32] and Roskam [33]. Calculations include a main and diversion mission. All following results refer to main mission outcomes; however, inclusion of reserves and diversion mission in the design loop is fundamental as it impacts aircraft weight and aerodynamics. Cruise performance is calculated in a discretized manner to account for on-flight fuel consumption, gradual reduction of carried weight, and therefore variation of thrust requirement. An estimation of takeoff field length and time to height are included in the mission calculation module. The aircraft dimensioning and weight calculations are coupled with the mission analysis and rubberized wing model to ensure the convergence of the needed fuel calculation, aircraft aerodynamics, and final maximum takeoff weight. By doing so, first-order effects on the aircraft weight and wing design are incorporated into the final outcome. It is important to note that while the battery system is designed and its performance is mapped by the electrical power system module, mission calculations are necessary to estimate the actual electrical energy consumption, given that cruise and other mission phases duration are calculated at that instance. Therefore, an energy flow calculation module is integrated at the end of the framework in order to analyze the share of each energy source for powering the propulsors and calculate battery depletion.

The last part of the integrated framework process is calculation of gaseous emissions. CO_2 and NO_x emissions are accounted for. CO_2 gaseous emissions are directly related to fuel burnt. Same goes for NO_x emissions, for which a semi-empirical correlation is used, derived by Kyprianidis et al. [10,40] for modern rich-burn quick-quench lean-burn (RQL) single-annular combustor designs. The correlation is based on simulated data and corresponding NO_x emissions measurements from the International Civil Aviation Organization (ICAO) engine emissions databank.

The integrated multidisciplinary framework that has been developed and employed combines all modules described in Sec. 2. A flowchart of its workflow is presented in Fig. 2.

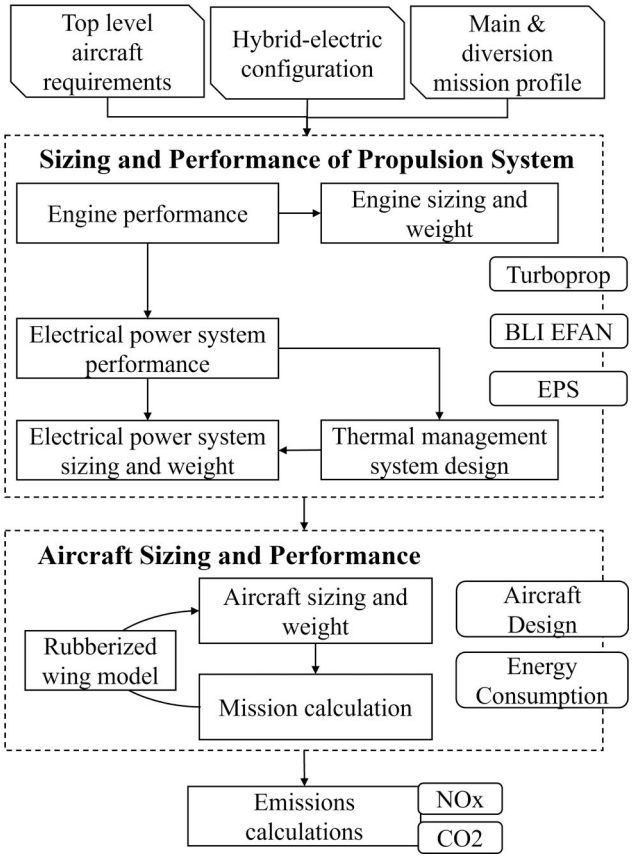


Fig. 2 Integrated framework for aircraft and power system design and performance

2.4 Reference Configurations. Three reference configurations are designed in this study in order to connect currently available and flying aircraft with the projected advances in technology and future aviation goals. Given that many of the current environmental objectives of the aviation industry have been set during the previous two decades, an aircraft with EIS 2014 is set as the primary reference case. This is a virtual airplane, matching state-of-the-art technology of that period. The EIS 2035 conventional aircraft is designed to verify and isolate the impact of technological improvements for conventional aircraft during this 20-year period. Finally, the hybrid EIS 2035 design brings perspective to the impact of electrification compared to an advanced conventional design. It also acts as the starting point for any following investigations regarding hybrid aircraft in this work. Validation of the reference aircraft is presented by Romano et al. [41] and extended by the HECARRUS consortium [36,39,42].

Top-level aircraft requirements and the most significant design aspects of these configurations are summarized in Table 3. Top-level and customer requirements are selected according to Romano et al. [41]. The one engine inoperative (OEI) scenario refers to failure of one turboprop engine. For the hybrid variants, nominal EFAN operation is assumed during OEI. Different multipoint synthesis schemes are used for the EIS 2014 and 2035 designs according to Sec. 2.1.

Moreover, the GT cycle of the EIS 2035 configurations is designed to be hotter and with a higher overall pressure ratio due to material advancements. Improvements in turbomachinery components are captured through polytropic efficiency values [31,43]. In terms of aircraft design, incorporation of composite materials in the frame and structure translates to reduced mass [44,45]. Additional structural integrity due to those materials also increases maximum allowed wing loading. Aspect ratio is kept constant among reference designs to allow for consistent comparisons. Representative aircraft power requirements capture the impact of MTOW along with the addition of the electrical power system for the hybrid configuration. For the nominal hybrid electric configuration of EIS 2035, an

Table 3 Design parameters and top-level requirements

		Conventional		Hybrid
	Units	EIS 2014	EIS 2035	EIS 2035
Top level requirements				
CS-23 MTOW limit	(kg)		8618	
Design mission range	(NMI)		400	
Max TOFL (SL, ISA)	(m)		800	
Min climb rate (SL, ISA, AEO)	(ft/min)		2150	
Cruise speed	(Mach)		0.35	
Passenger capacity			19	
Thermal powerplant				
OPR at CR		8.2	15	
T4 at CR	(K)	1140	1400	
GGT NGV Tm at TO	(K)	1140	1200	
Propeller power at TOC	(kW)	850	800	
Propeller power at TO	(kW)	1000	900	
GGC polytropic efficiency	Abs. Δ(%)	Reference	+2%	
GGT polytropic efficiency	Abs. Δ(%)	Reference	+0.5%	
FPT polytropic efficiency	Abs. Δ(%)	Reference	+0.5%	
EFAN PR at TOC		—		1.25
Electrical powerplant				
Cell specific energy (uninstalled)	(kWh/kg)	—		0.65
Cell specific power (uninstalled)	(kW/kg)	—		0.6
System voltage	(V)	—		515
El. drive design power	(kW)	—		460
Battery design power	(kW)	—		470
Aircraft				
Wing aspect ratio			11	
Max wing loading	(kg/m^2)	200	350	350
Materials		Conventional		Composite
Aircraft power requirement at TOC	(kW)	1700	1600	2000
Aircraft power requirement at TO	(kW)	2000	1800	2200

Table 4 Performance metrics for reference configurations

		Conventional		Hybrid
	Units	EIS 2014	EIS 2035	EIS 2035
Top level requirements				
MTOW	(%)	7865 (kg)	−17.0	+8.0
TOFL (SL, ISA)	(%)	715 (m)	+11.2	+9.1
Climb rate (SL, ISA, AEO)	(%)	2190 (ft/min)	+13.7	−1.8
Mission performance				
Block fuel	(%)	Reference	−28.6	−17.9
Block el. energy	(kWh)	—		400
Block energy	(%)	Reference	−28.6	−18.2
Block CO_2	(%)	Reference	−28.2	−17.8
Block NO_x	(%)	Reference	−34.4	−25.1
Thermal powerplant				
PWSFC at CR	(%)	Reference	−20.1	−27.9
PWSFC at TOC	(%)	Reference	−16.0	−32.3
Propeller loading at CR	(%)	Reference	−29.2	−58.8
Dry mass GT	(%)	Reference	−36.1	−33.3
Dry mass propeller	(%)	Reference	−21.9	−9.4
Electrical powerplant				
Battery mass	(kg)	—		920
Battery energy	(kWh)	—		520
Battery energy density (installed)	(kWh/kg)	—		0.56
Total EPS mass	(kg)	—		1180
Motor drive specific power	(kW/kg)	—		5.6
Aircraft				
Wing loading	(%)	200 (kW/m^2)	+27.5	+15

optimistic assumption of Li-ion battery technology is made. Design power for the battery system and EFAN electrical drive emerges from aircraft power requirements, power split between propulsors, and transmission efficiency of electrical branches.

Key performance metrics for these three reference configurations are presented in Table 4. Improvement in materials technology leads to a lighter conventional EIS 2035 aircraft compared to its EIS 2014 counterpart. Despite that, the hybrid aircraft carries significant battery mass resulting in increased MTOW, but still within certification limits. The wing is designed, among other aspects, in order to satisfy the takeoff and climb performance top-level requirements. The increased EIS 2035 margin in wing loading allows for the conventional and hybrid EIS 2035 wings to be designed with smaller surface compared to the EIS 2014 reference configuration. Along with the MTOW variation and the derived wing, wing loading increases by 27.5% and 15% for the conventional and hybrid EIS 2035 configurations, respectively.

Specific fuel consumption benefits in the order of 20% and 28% are realized for the conventional and hybrid EIS 2035 aircraft, respectively. This reduction is driven by the increase in turbomachinery efficiency and the hotter and higher OPR cycle that these are designed for. The hybrid aircraft gains an additional 8% improvement of specific fuel consumption compared to its conventional counterpart due to the addition of the battery-powered driven aft fan, while the gas turbine design power conditions stay constant. Propeller loading reduces for the EIS 2035 designs and significantly more for the hybrid version, due to different sizes. Finally, thermal powerplant mass reduces for future designs, and while gas turbine weights are close for the conventional and hybrid EIS 2035 designs, propeller mass for the hybrid aircraft increases compared to the conventional. In terms of mission outcomes, the gas turbine performance and respective change in MTOW along with the constant mission design range lead to block consumption and emissions reduction for the EIS 2035 designs compared to the conventional EIS 2014 aircraft. However, the reference hybrid aircraft underperforms compared to its conventional counterpart. Despite the specific fuel consumption benefits for the hybrid, the increase in MTOW due to the added battery and EPS mass results in increased power and thrust requirements during cruise. Hence, block fuel consumption increases for the hybrid reference aircraft compared to the conventional configuration. Block emissions closely follow these trends as they are much dependent on conventional fuel consumption, with overall improvements compared to 2014 in the order of 30%, but with the hybrid EIS 2035 reference case yielding benefits 10% lower compared to the conventional EIS 2035 aircraft. Therefore, environmental performance of the reference hybrid EIS 2035 design ends up being worse compared to the conventional EIS 2035 aircraft.

For the hybrid configuration, battery mass contributes to 78% of the overall electrical power system, which in turn is about 14% of maximum takeoff weight. Hence, to achieve around 20% degree of power hybridization, the resulting battery mass weighs twice as much as the consumed block fuel. The uninstalled (cell-level) battery energy density is 0.65 kWh/kg for the assumed EIS 2035 technology. Accounting for the packaging and auxiliary components, the installed battery system specific energy comes to 0.56 kWh/kg which results in a packaging factor of 13%. Finally, the resulting design of the electrical drive that is coupled to the EFAN has a specific power of 5.6 kW/kg.

By assessing the relative change in performance metrics between the conventional and hybrid EIS 2035 designs, it becomes apparent that optimal power management strategies and proper employment of the electrical power system are necessary for the latter to become competitive. Careful exploration of the design space and effective utilization of the integrated power system are crucial in exploiting the full potential of any complex hybrid architecture. Therefore, the optimal operation and design of the hybrid-electric architecture are explored in Sec. 3.1, in an effort to extract the outmost benefits of the configuration.

3 Results and Discussion

A design space is defined with the gas turbine and EFAN power settings being the main variables of interest in the pursuit of guidelines for optimal power management strategies. Around 5000 points have been selected with the Latin hypercube sampling method and simulated. The given method has been chosen for design space coverage.

The mission profile simulated comprises of takeoff, climb, cruise, descent, approach, landing, and taxi. Cruise Mach number is 0.35. The minimum constraint on climb rate is 2150 ft/min [41], while the max continuous climb rating is such that a 4% climb gradient at cutback point during the takeoff process is provided. Diversion

Table 5 Power setting and degrees of hybridization for the design space

	Lower bound	Reference	Upper bound
	EFAN power (source: battery) (kW)		
Takeoff	220	420	660
Climb	200	400	600
Cruise	50	200	350
	Aircraft power requirement (kW)		
Takeoff		2200	
Climb		2000	
Cruise		1000	
	Degree of hybridization for power (%)		
Takeoff	10	19	30
Climb	10	20	30
Cruise	5	20	35

mission range is 100 nautical miles, while the hold requirement is for 30 min on cruise power requirements. The detailed main and diversion mission envelope has been presented by Bermperis et al. [11]. Three mission segments are identified for having major impact in performance metrics of interest. These are takeoff, climb, and cruise.

Examined ranges and corresponding degrees of hybridization during those mission segments are presented in Table 5. Degree of hybridization is defined as the ratio between electrical power delivered to EFAN from the battery over the total aircraft power requirement. Selected range limits are indicative of design feasibility for the thermal and electrical components. Crossing the suggested margins poses design and sizing constraints for the thermal powerplant and subsequently for the electrical drives that are coupled to them. Moreover, electrical and thermal components may underperform, and further operability challenges arise. For the assumed battery, degrees of hybridization above the suggested lead to prohibitive battery masses, or subpar battery utilization during flight.

Reference values of power settings for all mission segments are presented in Fig. 3. Power levels for the EFAN and gas turbines during mission segments that are not included in Table 5 are kept constant during the evaluation of power management strategies. Takeoff and climb aircraft power requirements, which are used also for the thermal powerplant design, are fixed, except if stated otherwise. The actual aircraft power and powerplant loadings at cruise are generated according to the aircraft design and mission requirements. EFAN and propeller shaft contribute to meeting power requirements. EFAN power levels shown in Fig. 3 correspond to input from both chemical and electrical energy sources, while the propeller is driven purely with chemical energy. The power output of the gas turbines refers to the power that drives the propellers and the part that assists in driving the aft fan through the electrical power system in partial turboelectric mode. The battery system outputs adequate power to drive the EFAN, accounting for the gas turbines' share in partial turboelectric mode and the electrical efficiency of all intermediate components.

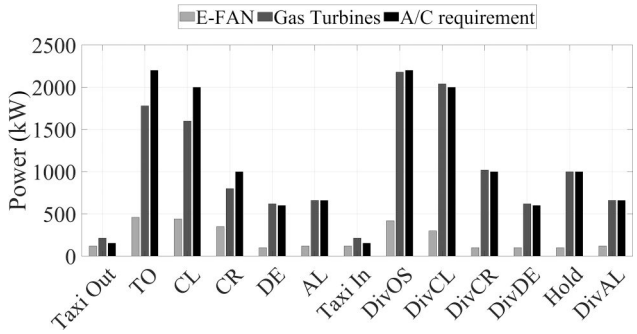


Fig. 3 Representative power levels for mission segments

Degree of hybridization in all segments apart from takeoff, climb, and cruise is at 0%, in order to isolate hybridization effects of mission parts that contribute the most to consumption and emissions. As Fig. 3 verifies, EFAN is always providing some propulsive power, as decoupling it from the energy sources and electrical drives has not been considered as an option. In mission segments that powering the EFAN via the battery is not desirable, the EFAN is powered via the gas turbines through the electrical branches in fully turboelectric mode.

3.1 Power Management Strategy for Entry-Into-Service 2035 Designs. An optimal power management strategy is one that results in minimum block fuel and energy consumption. It should also yield minimal emissions during flight. This study considers CO_2 and NO_x emissions. For a design to be acceptable, top-level and customer requirements need to be met. The maximum takeoff weight of a certification specifications (CS-23) aircraft is 8618 kg. Takeoff and climb performance are also monitored through takeoff field length and rate of climb, respectively. Finally, the lower bound of the end-of-mission battery state of charge is set at 20% to ensure slow degradation and allow for reserved energy in emergency scenarios. Given the complexity of the design space and number of constraints imposed on the problem, a set of relaxed constraints is also defined, and the subsequent filter is employed in the investigation. The initially selected and relaxed limits are presented in Table 6. The purpose of this is to evaluate the trends of performance metrics at their completeness and assess whether despite the adverse effect of battery addition, there is further untapped space to capitalize on electrification benefits.

During the following investigation, battery technology is kept fixed and refers to an optimistic projection for year of EIS 2035. Specific energy and specific power at cell level (uninstalled) are set at 0.65 kWh/kg and 0.6 kW/kg, respectively, while design system voltage is at 515 V. The electrical and thermal power systems are designed according to the combination of key power settings for each case of the design space. Design of the aircraft fuselage for the EIS 2035 hybrid aircraft group is fixed, while design of wings and tailplane vary according to the suggested wing loading of Table 4 and in order to achieve a reasonable center of gravity. Mission range and passenger capacity are also fixed according to the presented top-level aircraft requirements (TLAR). All results that are presented in a relative manner during this investigation are compared to the reference hybrid EIS 2035 aircraft, whose design details are presented in Tables 3 and 4 and whose reference hybridization values are shown in Table 5.

A multitude of points from the design space are simulated, each representing a different aircraft design. From around 5000 different designs, the best performing ones in terms of block fuel and energy consumption are collected and presented in comparison with the reference hybrid design in Fig. 4. For this case, best performing designs are deemed those that achieve the greatest savings for given power settings, while depleting the designed battery fully, irrespective of size. Each of the two graphs contains one set of curves for the initial constraints and one for the relaxed ones of Table 6. The best performing designs constrained with the initial limits reveal that acceptable ones reach EFAN power of 460 kW, 440 kW, and 365 kW, which correspond to degrees of hybridization

Table 6 Design space exploration set and relaxed constraints of top-level requirements

	Constraint	Relaxed constraint
SOC (%)	≥20	≥15
MTOW (kg)	≤8618	≤9050
RoC (ft/min)	≥2150	≥2000
TOFL (m)	≤800	≤1000

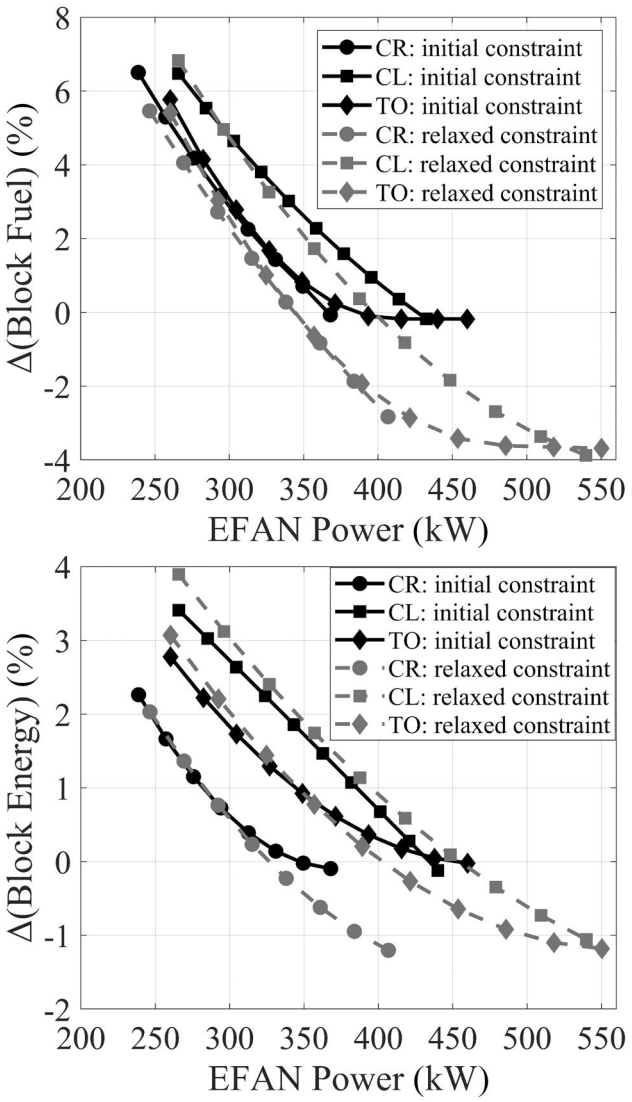


Fig. 4 Block fuel (top) and block energy (bottom) consumption of best performing designs for initial and relaxed constraints with respect to EFAN power

of 19%, 20%, and 20% for takeoff, climb, and cruise, respectively. Despite the use of twice as light batteries in designs of low degree of hybridization (5–10%), neither block energy nor block fuel reduction is possible. Less battery capacity directly translates to less electrification and a greater share of chemical energy in the overall mix. Less electrification also implies greater dependency on the thermal powerplant which operates with significantly less efficiency compared to the electrical power system, resulting in an overall increase of required block energy in spite of a lighter aircraft. Such designs of lower hybridization result in fuel penalties of up to 7% and energy penalties of up to 4%, with respect to the reference design.

Yet even the best emerging power management strategies do not yield further fuel and energy savings compared to the reference hybrid design. Relaxed constraints allow for acceptable designs of up to 550 kW, 550 kW, and 425 kW of EFAN power, corresponding to degrees of hybridization of 23%, 25.5%, and 27.5% for takeoff, climb, and cruise, respectively. Those designs yield a further 3–4% reduction of block fuel and 1% reduction of block energy compared to the reference hybrid case. This trend implies the suitability of the examined hybrid concept for bigger aircraft, such as the regional class.

Comparing trendlines for the EFAN power and hybridization of different mission segments, it occurs that cruise has greater impact in savings for the same power level compared to climb and takeoff. In relative terms, however, the same power level in all segments translates to greater hybridization for cruise. This implies that for the commuter aircraft, even though its cruise phase is smaller compared to larger aircraft, it is still electrification of this segment that holds the greatest potential for environmental benefits.

One of the main constraints of the evaluation is aircraft MTOW. Relaxing what is deemed acceptable in terms of aircraft mass by 5% and having a vessel of a bit over 9000 kg allows for more electrical energy to be installed. Heavier aircraft may violate takeoff and climb performance limitations which implies that designs of even lower wing loading may be needed to facilitate customer requirements, or that simultaneous relaxation of constraints is necessary as these are intertwined.

Battery mass and power settings of interest are correlated in Fig. 5, and bounds in the areas of 1000 kg and 1300 kg for the regular and relaxed limits, respectively, are revealed. Figure 5 also suggests that it is both power density and energy density of the battery that are of importance and critical for the design, as neither of the selected mission segments dominates battery mass trends. Finally, a filter in terms of the level of battery depletion is activated on the simulated cases in Fig. 5, highlighting the ratio of EFAN power at takeoff and cruise that results in the greatest exploitation of the installed electrical energy.

To further highlight the importance of a fully depleted battery and connect metrics of interest, Fig. 6 relates the differences in maximum takeoff weight and block fuel consumption. All simulated points are presented, and a filter indicates whether they are acceptable designs and under which constraints. Maximum takeoff weight and block fuel consumption constitute competing objectives, and a Pareto front from the best performing designs is constructed and presented in Fig. 6. It is no coincidence that all configurations that reach the prescribed limit of end-of-mission battery state of current (SOC) lie near the Pareto front. Moreover, it is verified once again that with the given technology assumptions and top-level aircraft requirements, it is very difficult to achieve further consumption benefits from what the reference hybrid design provides.

Apart from the effect of MTOW, rest customer requirements, battery system design and levels of its depletion, performance of the individual powerplants, and integrated power system also impact the optimal power management strategy. Energy and fuel consumption during cruise are an order of magnitude larger than any other individual mission segment. Therefore, cruise performance of the gas turbines, EFAN, and electrical branches is of great importance and calls for further investigation. Power system efficiency is shown

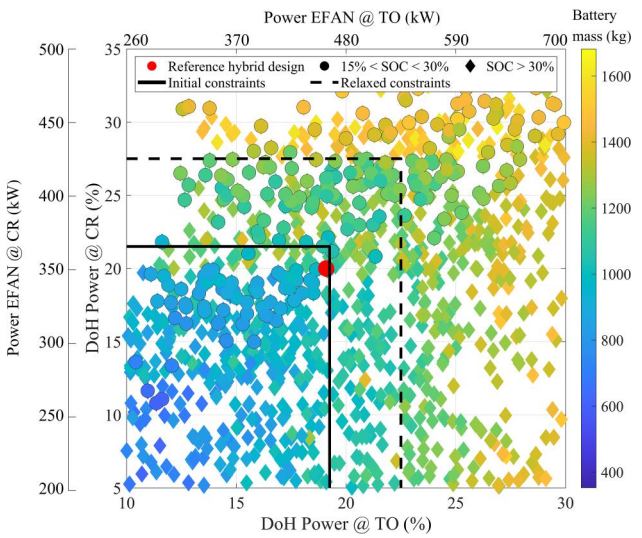


Fig. 5 Battery mass variation with cruise and takeoff power and degrees of hybridization

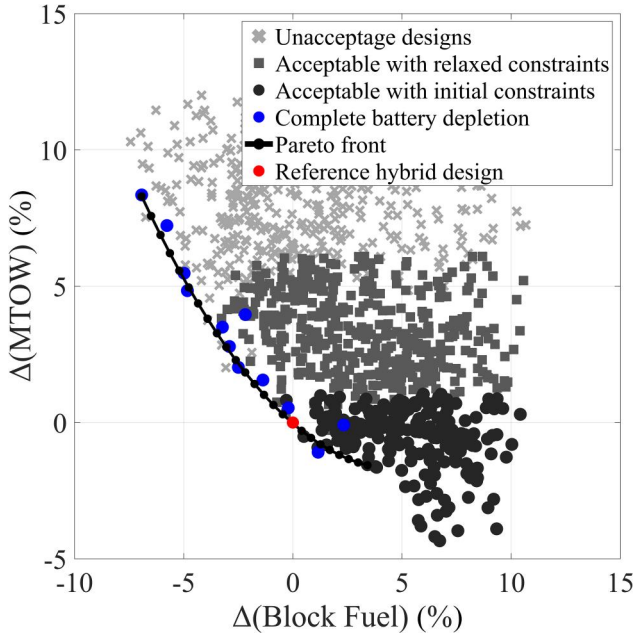


Fig. 6 Pareto front between MTOW and block fuel consumption for the investigated objective space

in Fig. 7 and is related to the selected power settings. A given EFAN power at a certain mission segment directly implies a propeller power requirement and therefore gas turbine power output. At the same time, electrical branch design and operational power is an outcome of power distribution between propulsors and degree of hybridization. Hence, not only all propulsors and power systems are interrelated and designed codependently, but they also depend on the combination of power levels of the mission phases of interest. This is enforced by the employment of a multipoint synthesis scheme for the thermal powerplant and the mission profile dependent design of the electrical power system.

The overall performance of the gas turbines and electrically driven aft fan is presented in Fig. 7. For the combination of EFAN power at cruise and climb, the upper left corner of the design space leads to nonfeasible thermal powerplants. Low EFAN design power at top of climb along with high power demand during cruise results in overspeeding conditions for the EFAN. Graphs on the left and middle of Fig. 7 showcase the difference between the simulated cases and the reference hybrid design in absolute percentage units. For the thermal power systems, EFAN power at cruise and climb show greatest impact on overall efficiency due to their role on the multipoint synthesis scheme. As EFAN power at climb increases, design power at top of climb increases for the EFAN and decreases for the gas turbines, due to the constant aircraft power requirement at that point. Increasing EFAN power at cruise corresponds to decreasing gas turbine power at the same phase.

In terms of gas turbine overall efficiency, improvements emerge in the area of increased climb EFAN power and decreased cruise EFAN power. This combination represents a lower power design gas turbine with higher relative loading during cruise. This design direction is beneficial for all individual performance components that comprise the overall efficiency, apart from propulsive, which is penalized for smaller size designs. The cumulative effect of core, installation, and transfer efficiency drives the outcome for gas turbines. A deviation of ±1.5% in overall gas turbine efficiency for the examined design space is revealed in the left graph of Fig. 7. For the EFAN, it is propulsive efficiency that drives the outcome of overall efficiency. As EFAN power at top of climb increases, the aft fan is designed larger with a positive effect on propulsive efficiency. A decrease in cruise EFAN power translates to lower fan pressure ratio for constant EFAN design, which again leads to better propulsive and overall EFAN efficiency. A deviation of ±3% is revealed in the middle graph of Fig. 7. It is worth noting that installation and transfer efficiency improve with an opposite trend compared to overall efficiency, but the impact of propulsive efficiency dominates the outcome for EFAN performance. Increased EFAN power during takeoff is also beneficial for overall efficiency of both thermal power systems, but its impact is less than of the shown power settings.

Therefore, to maximize thermal powerplant efficiency, the designer of these systems would aim to achieve a high degree of hybridization on the design point and combine it with low degree of hybridization during cruise. This indicates a large EFAN and smaller gas turbines and propellers, with relatively higher cruise loading for the gas turbines and lower for the EFAN.

For the electrical branch, cruise performance depends on the relation between branch design power and loading. The electric motor impacts branch performance the most. Optimal performance is achieved in the region of medium to low cruise loading with respect to the design point that usually comes from the takeoff power setting, given that EFAN and aircraft power are greater there. However, as indicated in Fig. 5, this region of the design space mainly involves configurations where batteries are not fully depleted. Therefore, the objective of the electrical power system designer becomes to select such a loading during cruise that will fully deplete the designed battery in the most efficient manner. This is achieved by having a ratio of at least 1:1 between cruise and takeoff degree of hybridization and an average minimum of 1:1.35 between cruise and takeoff EFAN power for the examined design space, with potential of further increasing cruise hybridization. This is the marginal line of acceptable hybrid designs and the space of optimal load and design power settings for the electrical power system. Efficiency in that area of optimality is in the range of 92–94%. The correlation of climb and cruise EFAN power reveals a

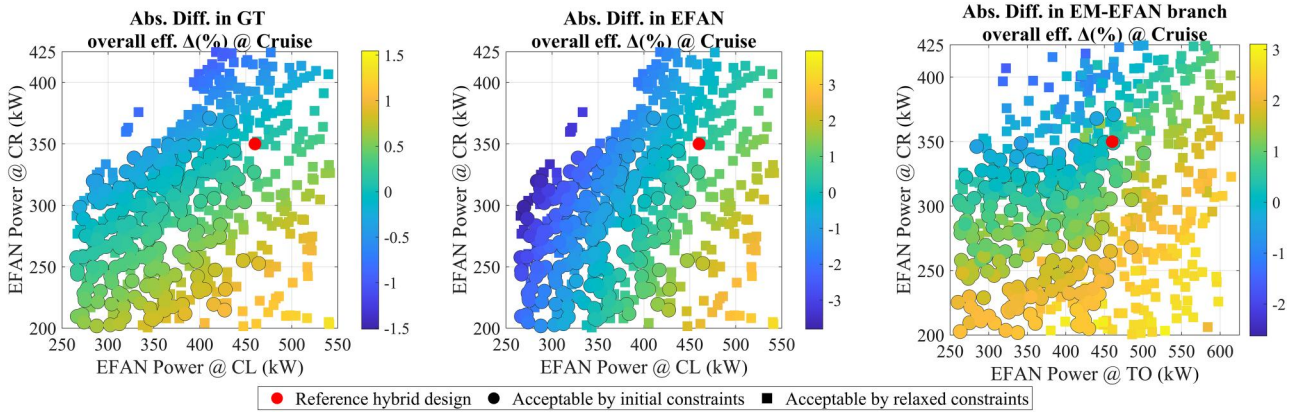


Fig. 7 Change in thermal and electrical powerplant performance at cruise with hybridization power

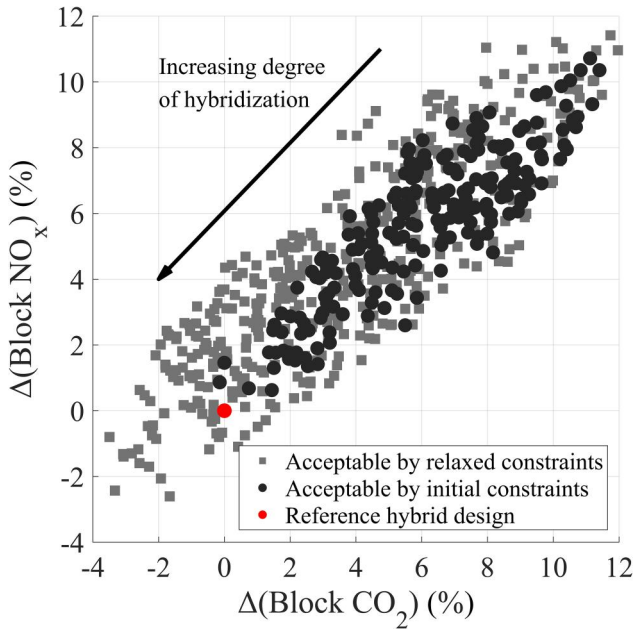


Fig. 8 Relative difference of block CO_2 and NO_x emissions for simulated cases

similar trend which is slightly dampened in terms of impact due to the stronger connection between EFAN takeoff power and the design point of the electrical branch.

Analysis of the individual power systems indicates a common area of performance optimality which is defined by high takeoff and climb EFAN power and low cruise EFAN power. In terms of battery depletion, however, this area corresponds to suboptimal electrical energy utilization and results in large battery designs where cruise electrification is not enough to take advantage of stored energy. As indicated by the curves of Fig. 4, such a power management strategy is not optimal and does not yield any further consumption benefits. For optimal energy utilization and maximization of consumption benefits, both the designer of the thermal and electrical power system should aim for as much electrification as possible in all mission segments. Achievable hybridization is limited by the selected customer requirements and top-level constraints, and the overall impact of increasing battery mass on aircraft performance. Despite this strategy not being optimal in terms of individual system performance, electrification results in enough fuel and energy savings to make it the overall best power management scheme for the examined space, aircraft architecture, and mission profile. Hence, the designer of a commuter series/parallel partial hybrid electric aircraft should strive for as much hybridization as possible, while having a cruise to takeoff or climb hybridization power ratio of at least 1:1.35, with the potential of increasing cruise power even further for given power settings in the other segments. This corresponds to a marginal ratio in degrees of hybridization between cruise and takeoff or climb of 1:1.

The objective space and trends for emissions of CO_2 and NO_x are presented in Fig. 8. Relative differences in block CO_2 and NO_x emissions compared to the reference hybrid design are shown. Both emission gases are closely related to block fuel consumption; therefore, increased hybridization, especially during cruise which contributes the most to consumption, is a driving factor for the revealed trend. For designs of lower degree of hybridization compared to the reference case, emissions of CO_2 and NO_x increase up to 10%. On the other hand, the relaxed design space allows for greater degrees of hybridization and reduced fuel consumption; therefore, further emissions reduction up to 3% are achievable. It is worth mentioning that designs that lead to a relative increase in both gaseous emissions above 8% have only suboptimal battery depletion; therefore, they do not capitalize on the carried electrical energy leading to greater environmental impact. While both types of emissions are well linked to cruise duration and performance, NO_x emissions are also strongly associated with combustor temperature. Hence, they depend on EFAN power during climb and subsequently top of climb which impacts cycle design and temperature. This relation is dampened for CO_2 emissions due to the stronger link with fuel consumption on cruise, and therefore EFAN power at that segment. This interrelation between emissions, power system design, and power management leads to the observed spread around the 1:1 ratio of emission deltas in Fig. 8.

3.2 Sensitivity Analysis. A sensitivity analysis with the purpose of generating exchange rates for the impact of the battery system assumptions on the hybridization of the hybrid-electric aircraft configuration and its environmental performance is conducted. The following exercise refers to the reference hybrid-electric aircraft of EIS 2035 that was presented in Secs. 2 and 3.

The driving parameter for the sensitivity analysis is installed battery energy density, which is varied by −1% in an attempt to showcase the results' behavior for worse assumed EIS 2035 battery technology. Two concepts are investigated along with the energy density change to capture the potential alternative paths that the hybrid-electric aircraft designer can follow. In both cases, the 1% energy density variation corresponds to a respective decrease of the carried electrical energy. If no further changes were made to the system design, the end-of-mission battery SOC would reduce below the selected value. To ensure that SOC remains same after the energy density change, the dark gray bars explore the necessary reduction of hybridization at cruise through the EFAN power, while keeping the battery system mass fixed. On the other hand, the light gray bars explore the increase of battery system mass so that the SOC remains same as the reference case and apply no variation on the hybridization power.

The outcomes of this analysis are showcased in Fig. 9. The decrease of installed battery energy density and simultaneous reduction of EFAN power at cruise results in the respective drop in the degree of hybridization. Block fuel consumption increases which leads to an increase in block energy. MTOW, takeoff, and climb performance show minimal variation, given that the battery mass did not change. NO_x and CO_2 emissions follow the variation of block fuel.

On the alternative path, the reduction of installed battery energy density is applied along with a respective increase of battery mass. Degree of hybridization is only slightly impacted due to the change

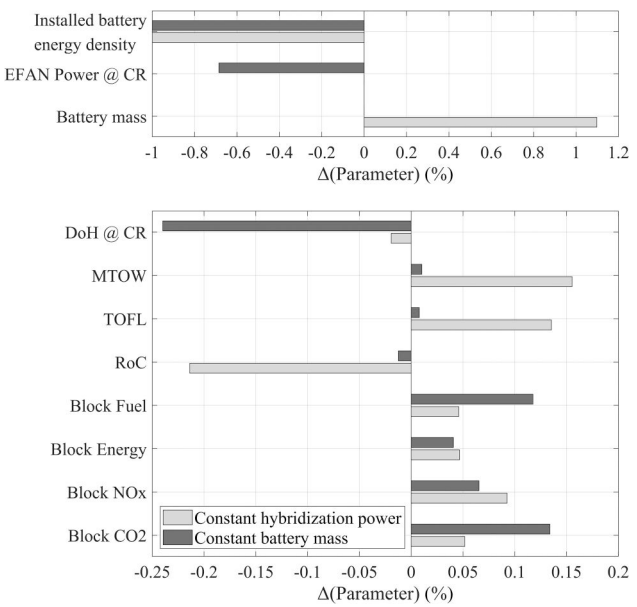


Fig. 9 Sensitivity analysis of installed battery energy density

in aircraft power requirement. However, the impact of battery on MTOW, takeoff, and climb performance is significant. Given the increased aircraft power requirement, block fuel and energy consumption increase. While the latter is comparable for the two concepts, block fuel penalties are greater when hybridization is reduced compared to when the resulting aircraft becomes heavier due to the added battery mass. This is also showcased in the variation of CO_2 emissions. However, for NO_x, other parameters apart from burnt fuel impact the result.

A 30% reduction on the reference installed energy density leads to a value of 0.4 kWh/kg for EIS 2035 battery technology. A hybrid aircraft with such energy density and of either of the examined paths performs worse than the reference hybrid configuration. Namely, the path of increasing the battery mass results in a design that does not satisfy the TLARs and initial constraints anymore (apart from SOC). Moreover, it yields an approximate further environmental performance penalty of only 1.5% compared to the original hybrid reference. On the other hand, the path of reduced hybridization is acceptable under the initial constraints, but fuel and CO_2 penalties increase by another 4%. The optimal alternative when addressing a lower installed battery energy density assumption would be to increase the battery system mass until the MTOW and aircraft performance limits are reached and only then reduce hybridization, sacrificing as less environmental performance as possible. Results presented in Fig. 9 agree with the individual outcomes of Sec. 3.1; therefore, the derived guidelines for optimal power management strategy of the selected architecture remain valid. The sensitivity analysis validates the notion that the examined hybrid electric architecture cannot perform better than the conventional one of the same EIS year.

3.3 Integrated Propulsion System and Aircraft Design. Through the studies thus far, derived guidelines on the effects of hybridization on propulsion system performance and mission-level outcomes have been discussed. Hence, the key design variables are degrees of hybridization through EFAN power at impactful design points and mission phases. However, in doing so, aircraft performance requirement constraints, such as takeoff, climb, and descent performance, are treated as bounds that confine the space of acceptable designs and not as fixed requirements that all aircraft designs need to abide to.

Designing the aircraft and propulsion system in an integrated manner with both powertrain and aircraft performance in mind, under fixed top-level aircraft requirements, is explored in Sec. 3.3 and compared with the initially investigated conceptual design study of Sec. 3.1. Takeoff field length, rate of climb, cruise speed and altitude, landing field distance, and descent time are the selected top-level aircraft requirements. The target values of those are selected to be equal to the reference hybrid EIS 2035 designs, for consistent comparison between derived hybrid aircraft. The same range of hybridization degrees presented in Table 5 is investigated. In order to achieve takeoff performance, power loading is varied. Rate of climb and time to height are coupled with the climb power requirement, while cruise speed and altitude are kept constant. Descent time is connected with descent engine ratings, and landing field distance is coupled with aircraft wing loading. Hence, through this integrated conceptual design approach, the propulsion system is electrified with selected degrees, and at the same time it is sized and operated according to top-level aircraft requirements.

The outcomes of this investigation are presented in Fig. 10. Results of the original conceptual design approach, initially presented in Fig. 6, are shown via the derived MTOW–block fuel optimal front, spanning −7.5% to 6% block fuel change. For the same degree of hybridization defined design space, the steeper curve showcases aircraft performance via a similar front of optimal designs, but for constant top-level aircraft requirements. Aircraft configurations that are lighter than the reference point due to less hybridization and lower battery mass perform better in terms of block fuel for the same derived aircraft mass. This is because the propulsion system design and operation conditions reduce to achieve the given target requirements, while the reference design

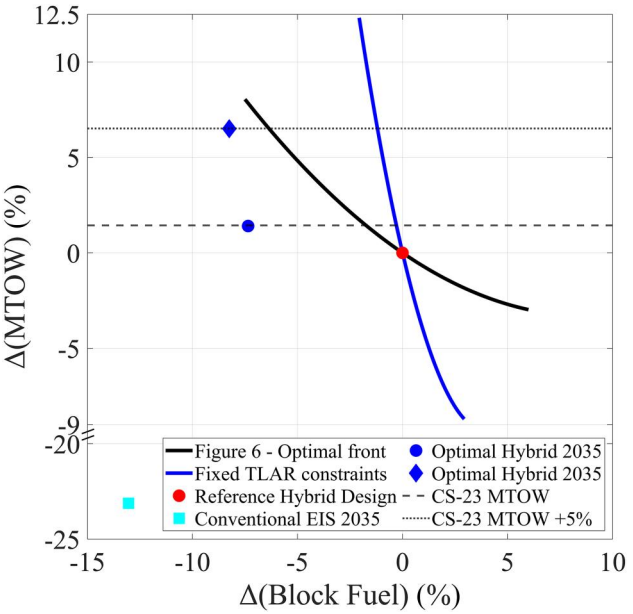


Fig. 10 Comparison of aircraft performance outcomes for propulsion system hybridization investigation under free and fixed aircraft top-level requirements

operates in more demanding conditions. On the other hand, configurations that are heavier due to more electrification need to have further uprated propulsion systems compared to the reference. This leads to greater fuel consumption for given installed battery and aircraft mass, compared to the original study. Greater electrification and battery masses require more powerful propulsion systems to carry the additional mass under fixed top-level requirement. This necessity neutralizes a lot of the initially found performance benefits; nevertheless, it does not overwrite the underlying propulsion system and hybridization trends, nor the guidelines for optimal power management.

Finally, Fig. 10 presents two optimally designed and operated hybrid electric EIS 2035 configurations. These lead to a further 7.5–8% reduction of block fuel and lie in the CS-23 MTOW limit and its relaxed version of +5% allowed mass. Thus, they maximize the installed electrical energy, which is proven as the most beneficial approach. The configuration with exactly the CS-23 aircraft mass is hybridized at 23.5% during takeoff and climb, and 26.5% at cruise. In order to allow for the originally explored design space of large variations in hybridization and propulsion system design, some suboptimal design choices in non-key parameters were made. This guaranteed that design space changes are applied only in key design parameters, without simulation issues. This was mainly applied in the diversion mission operation and gas turbine power off-take for operability in extreme cases of the design spaces. In the two optimal hybrid EIS 2035 designs, these are retracted, and all design choices are made with block fuel reduction in mind, while following the derived hybridization guidelines and adhering to the selected top-level aircraft requirements. However, the conventional EIS 2035 configuration still consumes 5% less fuel compared to the optimal hybrid designs.

4 Conclusion

Ambitious environmental objectives of aviation for the near future call for the investigation of novel electrified concepts. A 19-passenger commuter aircraft employing a series/parallel partial hybrid electric architecture with an electrically driven aft fan operating under boundary layer ingestion conditions is examined for its potential.

Design and analysis of a reference hybrid-electric aircraft with EIS 2035 has shown several environmental benefits compared to its

conventional counterpart of EIS 2014. Block fuel and energy consumption improve in the order of 18%, with gaseous NO_x and CO_2 emissions following the trend. In an effort to maximize such benefits even further, a design space exploration is performed with the objective of generating guidelines for the optimal power management strategy. The study verified that the added battery mass and its relation to the aircraft maximum takeoff weight play a limiting role on the degree of hybridization. Nonetheless, increased electrification may yield further consumption benefits if certain top-level aircraft requirements are relaxed. Degrees of hybridization up to 27.5% and added consumption benefits in the order of 5% from what the reference design already achieves are possible. Despite the fact that an area of individual power system performance optimality is found for small cruise degrees of hybridization and larger EPS and EFAN designs, these do not overlap with the aircraft level optimal. The best power management strategy is installing as much electrification as the constraints allow and ensuring that the battery is depleted to its allowed limit by the end of the mission. A minimum ratio of 1:1.35 in hybridization power between cruise and takeoff or climb realizes this condition. Battery systems of up to 1300 kg are designed in this process. The investigation revealed that CO_2 and NO_x trends follow closely fuel consumption, but also the design point of the thermal powerplant which is interrelated to the hybridization and power split between propulsors.

Despite all efforts to increase the benefits of the novel hybrid configuration, the explored hybrid designs result in worse environmental performance compared to the conventional counterpart of EIS 2035. Therefore, the explored hybrid architecture is deemed unsuitable for the 19-passenger commuter aircraft class. This outcome is further supported when less optimistic battery technology is assumed. A 30% decrease on the installed battery energy density corresponds to either a 33% increase of battery mass, or 7% decrease on the achievable degree of hybridization. In both cases, either aircraft or environmental performance are penalized. Nevertheless, the optimal power management guidelines may be applied to aircraft of larger classes, such as regional, as potential is indicated by the accumulated benefits of the exploration under relaxed constraints.

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

AC = alternating current
AEO = all engine operative
CL = climb
CLMAX = maximum lift coefficient
CR = cruise
CS = certification specifications
DC = direct current
Diff. = difference
DoH = degree of hybridization
EFAN = electric fan
EIS = entry-into-service
EM = electric motor
EoM = end of mission
EPS = electrical power system
EVA = Environmental Assessment Tool
FPT = free power turbine
GEN = generator
GGC = gas generator compressor
GGT = gas generator turbine
GT = gas turbine
ISA = International Standard Atmosphere
Li = lithium
MTOW = maximum takeoff weight
NGV = nozzle guide vane
OPR = overall pressure ratio
PWSFC = power specific fuel consumption
RoC = rate of climb
SiC = silicon carbide
SL = sea level
SOC = state of current
Tm = metal temperature
TO = takeoff
TOC = top of climb
TOFL = takeoff field length
T4 = turbine inlet temperature

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